Mirai Botnet Detection: ML with Live IoT Simulation

PROJECT REPORT HONOURS

Submitted by

Shegin Ismail A P (IDK20IT038)

in partial fulfillment for the award of degree of

Bachelor of Technology(Honours)

in

Information Technology

of

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

KERALA



DEPARTMENT OF INFORMATION TECHNOLOGY

GOVERNMENT ENGINEERING COLLEGE IDUKKI, PAINAVU - 685603

KERALA

May 2024

DEPARTMENT OF INFORMATION TECHNOLOGY

GOVERNMENT ENGINEERING COLLEGE IDUKKI, PAINAVU-685603

2023 - 24

CERTIFICATE

This is to certify that the project design document entitled **Mirai Botnet Detection: ML with Live IoT Simulation** is a bonafide record of the Project work presented by **Shegin Ismail A P** (IDK20IT038),in partial fulfillment of the requirements for the award of the Degree of **Bachelor of Technology(Honours)** in **Information Technology** of APJ Abdul Kalam Technological University, Kerala.

Prof. Sanu C
(Project Coordinator & Guide)
Assistant Professor
Dept. of IT, GECI

Dr. Remesh Babu

Head of the Department
Dept. of IT, GECI

DECLARATION

We hereby declare that the project report **Mirai Botnet Detection: ML with Live IoT Simulation**, being submitted in partial fulfillment for the award of B.Tech(Honours) degree is the original work carried out by us. It has not formed the part of any other thesis submitted for award of any degree or diploma, either in this or any other University.

Painavu
20-4-2024

Shegin Ismail A P (IDK20IT038)

ABSTRACT

The proposed project focuses on the development of a comprehensive Mirai IoT botnet detection system, utilizing a Python tool coupled with machine learning algorithms for real-time packet analysis and identification. The system is designed to monitor network traffic, capturing live packets and analyzing essential packet information. It specifically targets the detection of Mirai botnet activity by leveraging a trained machine learning model and explicitly identifying ports commonly associated with Mirai attacks.

Key functionalities of the system include live packet capture, packet analysis, and automated detection of Mirai botnet presence. Upon detection, the system triggers various actions, including email notifications with detailed packet summaries attached as PDF documents, updates to Firebase database for logging purposes, and activation of alerts such as turning off room lights and activating a buzzer in an IoT environment using an Android emulated Flutter mobile application developed for controlling room lights, offering real-time interaction with the IoT environment.

Furthermore, the system provides the flexibility to upload PCAP files for retrospective analysis of Mirai presence. the Mirai IoT botnet detection system offers a robust solution for proactive threat detection, providing real-time insights into network security and facilitating rapid response mechanisms to safeguard against Mirai botnet attacks in IoT environments.

ACKNOWLEDGEMENT

It is our privilege to express our sincere and deep gratitude and indebtedness to the people who helped us to complete the project successfully. All glory and honors to Almighty God, who showered His grace upon us to make this endeavor a success. We thank Dr. Suresh K., Principal, Government Engineering College Idukki, for providing new dimensions to this engineering course. We express our gratitude to Prof.Sanu C, project guide, Department of Information Technology, Government Engineering College Idukki, for his valuable suggestions, encouragement, and advice for the fruitful completion of this project.

We express our deep sense of gratitude to the project coordinator Prof. Sanu C Assistant Professor in the Department of Information Technology, Government Engineering College Idukki, for their valuable suggestions and review comments throughout the project work. Thanks to all the teaching and non-teaching staff of the Department of Information Technology, Government Engineering College Idukki, for their valuable support and help.

Shegin Ismail A P

Contents

ABSTRACT **i**

List of Tables **v**

List of Figures **vi**

ABBREVIATIONS **vii**

1 INTRODUCTION **1**

2 LITERATURE REVIEW **2**

2.1 LITERATURE SURVEY 2

2.1.1 The evolution of Mirai botnet scans over a six-year period . . 2

2.1.2 Intelligent Detection of IoT Botnets Using Machine Learning and Deep Learning 3

2.1.3 Packet analysis for network forensics: A comprehensive survey 4

2.1.4 Classification Based on Decision Tree Algorithm for Machine Learning 5

2.1.5 Google Firebase based Modern IoT System Architecture . . . 6

2.2 OVERVIEW AND OUTCOME OF LITERATURE SURVEY 8

3 PROBLEM DEFINITION **10**

3.1 PROBLEM STATEMENT 10

3.2 PROBLEM DESCRIPTION 10

3.3 SCOPE AND OBJECTIVES 11

3.3.1 SCOPE 11
3.3.2 OBJECTIVES 11

4 SYSTEM DEVELOPMENT 13
4.1 Software Requirements 13
4.1.1 Requirement Specification 13
4.1.2 Non-Functional Requirements 15
4.1.3 Hardware Requirements 16

5 SYSTEM DESIGN 17
5.1 System Architecture 17
5.2 Class Diagram 18
5.3 Use Case Diagram 19

6 IMPLEMENTATION 21
6.1 IMPLEMENTATION TOOLS 21
6.1.1 Mirai Detection Tools 21
6.1.2 Mobile App Development Tools 22
6.1.3 Hardware Components and Modules 23
6.2 RESULTS AND DISCUSSION 24
6.2.1 Algorithm Accuracy 24
6.2.2 Mirai Detection Tool 24
6.2.3 Realtime IoT App 26

7 TESTING 29
7.1 Functionality Testing 29
7.2 UI Testing 29
7.3 Integration Testing 30

8 CONCLUSION 31
8.1 Definitions 32

REFERENCES 34

List of Tables

2.1 Overview of Literature Survey 8
2.2 Outcome of Literature Survey 8

List of Figures

5.1 System Architecture 18
5.2 Class Diagram 19
5.3 Use Case Diagram 20

6.1 Accuracy of Decision Tree Classifier against other algorithms 24
6.2 Mirai Detection main page 25
6.3 Packet Analysis 25
6.4 Email Notification 26
6.5 Authentication Pages 27
6.6 Home Page 28
6.7 Circuit 28

ABBREVIATIONS

- **Wi-Fi**:Wireless Fidelity
- **IoT**:Internet of Things
- **ML**:Machine Learning
- **NodeMCU**: Node Microcontroller Unit
- **PCAP**: Packet Capture
- **PDF**: Portable Document Format.
- **DDoS** : Distributed denial-of service
- **IP**: Internet Protocol
- **IPV4**:Internet Protocol Version 4
- **IPV6**:Internet Protocol Version 6
- **TCP**:Transmission Control Protocol.
- **SMTP**:Simple Mail Transfer Protocol

CHAPTER 1

INTRODUCTION

The proliferation of Internet of Things (IoT) devices has significantly enhanced connectivity and convenience in various domains,. However, with this increased connectivity comes the risk of security threats, including the notorious Mirai botnet which is known for its large-scale distributed denial-of-service (DDoS) attacks, exploits vulnerable IoT devices, posing significant risks to network integrity and data security.This project presents a comprehensive Mirai IoT botnet detection system.Key components of the Mirai IoT botnet detection system include a Python-based tool for live packet capture and analysis, supplemented by a trained machine learning model tailored for Mirai detection.The system also facilitates retrospective analysis by allowing users to upload PCAP files for comprehensive examination of historical network traffic.The system seamlessly integrates with Firebase for real-time logging and reporting of Mirai detection events. It generates email notifications with detailed packet summaries as PDF documents upon Mirai detection, offering stakeholders timely alerts and insights. Additionally, it extends functionality to IoT environments, enabling control and monitoring of connected devices via an Android emulated Flutter mobile app.

CHAPTER 2

LITERATURE REVIEW

2.1 LITERATURE SURVEY

2.1.1 The evolution of Mirai botnet scans over a six-year period

The Mirai botnet, since its emergence, has posed a significant threat to the security of networked systems, orchestrating large-scale DDoS attacks and exploiting vulnerabilities in IoT devices. This paper by A.Affinito *et al.*[1] review synthesizes existing research on the evolution of Mirai botnet scans over a six-year period, encompassing detection techniques, mitigation strategies, and the analysis of scanning patterns.

Botnet Landscape and Historical Context

The prevalence of botnets, particularly since the early 2000s, has underscored their role as a primary source of Internet security failures. Various botnets, such as Hydra, Aidra, and BASHLITE, have been instrumental in executing spamming, phishing, and DDoS attacks. The historical context provided by previous research highlights the persistent and evolving nature of botnet threats, setting the stage for the emergence of Mirai as a seminal example of botnet innovation and adaptation.

Botnet Detection Techniques

Detection of botnet activities has been a focal point of research, with studies leveraging diverse methodologies to identify compromised devices and malicious behavior. Machine learning algorithms, in particular, have emerged as effective tools in the detection of botnets, leveraging features extracted from network traffic to identify intrusions and compromised IoT devices. Research efforts have explored techniques ranging from passive measurements to deep learning methodologies, highlighting the importance of adaptive detection mechanisms in combating evolving botnet threats.

Mirai Botnet Detection and Mitigation

The Mirai botnet, in particular, has been the subject of extensive research on detection and mitigation strategies. Adaptive network layer techniques, machine learning algorithms, and analysis of network traffic traces have been employed to enhance detection and response mechanisms. Research efforts have focused on understanding Mirai's behavior, propagation patterns, and target ports, providing insights into its evolution over time. The analysis of scanning patterns and target ports has informed strategies for detection and mitigation, emphasizing the need for adaptive approaches to combat Mirai and similar botnet threats effectively.

2.1.2 Intelligent Detection of IoT Botnets Using Machine Learning and Deep Learning

The paper by Kim *et al.*[2] provides a comprehensive review of recent advancements in IoT botnet detection methodologies, focusing primarily on machine learning (ML) and deep learning (DL) approaches. It synthesizes findings from studies in anomaly detection, malware classification, and network anomaly detection using datasets such as N-BaIoT. The paper proposes a robust framework outlining the process of developing botnet detection models, which are rigorously evaluated through binary and multiclass classification experiments. Results highlight the effectiveness of various ML and DL algorithms, particularly decision trees, random forests, and convolutional

neural networks (CNNs), in detecting IoT botnets across diverse device types. Furthermore, the study emphasizes the importance of sophisticated detection models to combat the evolving threats posed by IoT botnets. It concludes by advocating for further research and development of integrated IoT security frameworks to enhance the resilience of IoT ecosystems against increasingly sophisticated cyber threats.

2.1.3 Packet analysis for network forensics: A comprehensive survey

Tools and Techniques for Packet Analysis

The survey paper by L. F. Sikos *et al.*[3] extensively covers a wide array of tools and techniques available for packet analysis. Notable mentions include Wireshark, Snort, Tcpdump, and Suricata, each offering unique functionalities ranging from basic packet capture to deep packet inspection and intrusion detection/prevention. Specialized tools like EtherApe and Capsa provide visualizations and in-depth packet decoding capabilities, while cloud-based solutions like Amazon's VPC traffic mirroring address the need for packet analysis in cloud environments. The literature highlights the versatility and importance of these tools in network forensics investigations, enabling practitioners to capture, analyze, and reconstruct network traffic for investigative purposes.

Challenges and Future Directions

The survey identifies several challenges and future directions in packet analysis for network forensics. Privacy concerns and legal issues surrounding packet analysis, particularly in wireless networks and IoT devices, emerge as critical areas requiring further research and development. The growing complexity of network data, including encrypted traffic and the proliferation of IoT devices, presents challenges for traditional packet analyzers. Moreover, the emergence of machine learning-based approaches holds promise in addressing issues related to big network data and encrypted streams. As cloud-based services continue to gain traction, there is a need for packet analysis

solutions tailored to cloud environments, emphasizing the importance of research in this area. Overall, the literature underscores the importance of addressing these challenges to advance the field of network forensics and packet analysis.

Applications in Network Forensics

The survey highlights the diverse applications of packet analysis in network forensics. From reconstructing network activities and identifying malicious behavior to gathering evidence for legal proceedings, packet analysis tools play a pivotal role in forensic investigations. Tools like NetScout TruView and NetworkMiner offer advanced features for tracing abnormal user behavior and detecting operating systems and open ports, respectively. Moreover, purpose-built carvers like tcpflow and FileTSAR efficiently extract files from packet captures, aiding in the forensic analysis of network traffic. The literature underscores the importance of these tools in enabling investigators to analyze network traffic comprehensively and derive valuable insights for forensic investigations.

2.1.4 Classification Based on Decision Tree Algorithm for Machine Learning

The paper by B. Charbuty *et al.*[4] encompasses a comprehensive examination of recent advancements in decision tree (DT) algorithms across diverse domains, emphasizing their efficacy and applications. Notable studies from various fields showcase the versatility and effectiveness of DT algorithms, shedding light on their methodologies, datasets utilized, and outcomes achieved.One study focused on survival prediction in Locally Advanced Rectal Cancer (LARC), utilizing decision tree algorithms to analyze patient data and identify predictive variables for survival, facilitating informed clinical decision-making. Another study aimed to predict regular smoking time, employing Extreme Gradient Boosting (XGBoost) for prediction. The decision tree approach yielded higher accuracy compared to Random Forest. In a different domain, a machine learning model for spam detection was developed, where decision tree models demonstrated high accuracy (99.93%), outperforming

other algorithms.Addressing a challenge in medical imaging, a study tackled the task of optic disc segmentation in retinal fundus images. By utilizing decision tree classifiers for adaptive thresholding, robust segmentation results were achieved even in images with anomalies.

In the realm of cybersecurity, a study focused on Distributed Denial of Service (DDoS) attack detection. Comparing Decision Tree (DT) algorithms against KNN, DT outperformed KNN with an accuracy rate of 99.91%, highlighting its robustness in cybersecurity applications. Another study delved into abnormal red blood cell detection using image processing. The decision tree algorithm achieved an average accuracy of 89.31% in identifying irregular red blood cells. In the domain of spam detection, researchers aimed to enhance detection using a hybrid decision tree and genetic algorithm approach. By combining genetic algorithm with decision tree pruning, accuracy improved to 95.5% post feature extraction using PCA.In gene expression analysis for malaria detection, researchers compared KNN with DT, finding that KNN outperformed DT with an accuracy of 86.7%.These studies collectively underscore the versatility and effectiveness of decision tree algorithms across diverse domains, ranging from healthcare to cybersecurity and image processing. By leveraging decision trees, researchers have achieved significant advancements in predictive modeling and pattern recognition tasks, demonstrating their utility in real-world applications.

2.1.5 Google Firebase based Modern IoT System Architecture

The literature survey by B Padmaja *et al.*[5] delves into the rapidly evolving landscape of Internet of Things (IoT) technology, emphasizing its integration with other technologies like Big Data, Machine Learning, and Blockchain. The introduction outlines the components and architectural principles of IoT systems, highlighting the significance of designing efficient and scalable systems amidst various challenges.

Existing implementations and their limitations are discussed, showcasing the challenges faced in implementing scalable IoT systems. Various proposals and architectures are explored, from home automation to highly scalable systems, each

presenting its set of limitations and complexities.

Firebase, a platform developed by Google, emerges as a promising solution for simplifying server-side development and scalability challenges. However, literature on using Firebase for IoT systems is scarce, with few experimental implementations available.

The proposed work introduces an IoT system leveraging Firebase cloud services, particularly Firebase Realtime Database (RTDB) and Firestore, for efficient data storage and real-time communication between IoT devices and consumer portals. The implementation details illustrate the architecture, system flow, and interaction between components, showcasing the feasibility and effectiveness of the proposed solution.

Experimental results demonstrate the successful implementation of the IoT system, showcasing real-time data synchronization, device control, and alerting functionality. The conclusion underscores the abstraction of complex server tasks, scalability of the system, and potential for future extensions, particularly with the integration of machine learning using Firebase's evolving platform.

Overall, the literature survey underscores the significance of leveraging platforms like Firebase for simplifying the development and scalability challenges associated with IoT systems, paving the way for more efficient and sophisticated implementations in the future.

2.2 OVERVIEW AND OUTCOME OF LITERATURE SURVEY

Table 2.1: Overview of Literature Survey

Sl No	Paper	Key Idea	Algorithm/Technique Used
1	A. Affinito *et al.*[1]	Synthesizing research on the evolution of Mirai botnet scans, detection techniques, and mitigation strategies.	Machine learning algorithms, deep learning methodologies, analysis of network traffic patterns.
2	J. Kim *et al.*[2]	Review of recent advancements in IoT botnet detection, focusing on ML and DL approaches.	Decision trees, random forests, convolutional neural networks (CNNs).
3	L. F. Sikos *et al.*[3]	Comprehensive coverage of tools and techniques for packet analysis, challenges, and future directions.	Wireshark, Snort, Tcpdump, Suricata, EtherApe, Capsa, Amazon's VPC traffic mirroring.
4	B. Charbuty *et al.*[4]	Examination of recent advancements in decision tree algorithms across diverse domains.	Decision trees, Extreme Gradient Boosting (XGBoost), genetic algorithms, KNN.
5	B Padmaja *et al.*[5]	Exploration of IoT system architecture leveraging Firebase cloud services.	Firebase Realtime Database (RTDB), Firestore, machine learning integration.

Table 2.2: Outcome of Literature Survey

Sl No	Paper	Good Things About the Work	Drawback/Limitations
1	A. Affinito *et al.*[1]	Comprehensive synthesis of Mirai botnet evolution research.	Limited discussion on mitigation effectiveness.
2	J. Kim *et al.*[2]	Rigorous evaluation of ML and DL algorithms for IoT botnet detection.	Lack of real-world deployment evaluation.
3	L. F. Sikos *et al.*[3]	Extensive coverage of packet analysis tools and techniques.	Limited discussion on cloud-based packet analysis.
4	B. Charbuty *et al.*[4]	Showcase of decision tree effectiveness across diverse domains.	Lack of comparison with other machine learning algorithms.
5	B Padmaja *et al.*[5]	Simplification of IoT system development using Firebase cloud services.	Limited discussion on scalability challenges with Firebase.

The literature survey delves into a multifaceted landscape, spanning cybersecurity, machine learning, network forensics, and IoT system architecture. Insights from studies on the Mirai botnet evolution and intelligent detection of IoT botnets underline the persistent threat posed by malicious actors and the efficacy of machine learning and deep learning algorithms in identifying and mitigating botnet activities. However, these findings also underscore the necessity for further research to assess the real-world applicability and scalability of detection models. Similarly, the exploration of packet analysis tools and techniques in network forensics emphasizes the critical need for robust solutions to capture, analyze, and reconstruct network traffic, particularly in light of challenges posed by privacy, legal compliance, and the evolving complexity of network data. In the realm of machine learning, the versatility and effectiveness of decision tree algorithms across diverse domains, from healthcare to cybersecurity and image processing, are underscored. Yet, there remains a gap in comparative analysis with other machine learning algorithms, warranting further investigation to determine optimal approaches for specific tasks. Lastly, the examination of Google Firebase-based IoT system architecture presents a promising avenue for simplifying server-side development and addressing scalability challenges. While Firebase offers significant advantages such as real-time data synchronization and simplified development, scalability challenges and limited literature on its use in IoT systems underscore the need for further research and evaluation. In conclusion, the literature survey highlights the complexity and interconnectedness of cybersecurity and IoT system development, with ongoing research efforts aimed at enhancing detection and mitigation strategies, improving network forensics capabilities, leveraging machine learning for predictive modeling, and streamlining IoT system architecture. Continued collaboration between researchers, industry stakeholders, and policymakers is crucial to address emerging cyber threats and develop robust solutions for securing networked systems and IoT ecosystems.

CHAPTER 3

PROBLEM DEFINITION

3.1 PROBLEM STATEMENT

To develop a Tool for Mirai Botnet Detection System in realtime environment

3.2 PROBLEM DESCRIPTION

Mirai botnet presents a significant threat to IoT networks, exploiting vulnerabilities for widespread DDoS attacks. Detecting Mirai activity amidst legitimate IoT traffic requires advanced packet analysis and machine learning algorithms. Integrating detection with real-time notification and IoT device management is crucial for timely response. A comprehensive solution demands sophisticated detection mechanisms and effective mitigation strategies. Safeguarding IoT ecosystems against Mirai threats requires a multi-faceted approach combining detection, notification, and device management functionalities.

3.3 SCOPE AND OBJECTIVES

3.3.1 SCOPE

The Mirai botnet represents a significant cybersecurity challenge, exploiting vulnerabilities in IoT devices to launch large-scale attacks. Addressing this threat requires proactive detection mechanisms. This project aims to develop a comprehensive solution by creating a Python-based tool capable of real-time packet capture and analysis for Mirai detection. The scope encompasses analyzing packet metadata, such as source and destination IP addresses and ports, leveraging trained machine learning algorithms to identify Mirai activity patterns. The tool will offer email notifications upon Mirai detection, including detailed packet summaries and integration with Firebase for data storage and retrieval. Additionally, users can upload pcap files for historical analysis. Furthermore, the project involves simulating an IoT network environment to validate detection capabilities. It includes developing a Flutter-based mobile application for remote device management, with automated responses triggered upon Mirai detection, such as disabling compromised devices and issuing alerts within the app interface. This holistic approach aims to enhance network security by detecting and mitigating Mirai botnet threats effectively.The dataset used has following attributes :Botnet,Source IP,Source Port,Source IP AsnNr,Target IP,Target Port,Source IP Region,Source IP City,Source IP Latitude,Source IP Longitude,Threat Confidence,Threat Classification, Host Address,Prediction

3.3.2 OBJECTIVES

The specific objectives of the system are :

1. **Perform Live Network Analysis**: Develop a Python tool capable of capturing live network traffic and analyzing packet information such as source and destination IP addresses, as well as source and destination ports in real-time.

2. **Perform Network Analysis of PCAP Files**: Enable the tool to analyze packet capture (PCAP) files obtained from previous network captures, allowing for

retrospective analysis of network traffic.

3. **Create a Machine Learning Model for Mirai Botnet Detection**: Develop and train a machine learning algorithm specifically designed to detect the presence of the Mirai botnet within network traffic based on identified patterns and behaviors.

4. **Integrate ML model to Live Packet Analysis and PCAP File Analysis**: Incorporate the trained machine learning model into both live packet analysis and PCAP file analysis functionalities of the tool to automate the detection process.

5. **PDF Document Generation with Details of Network Analysis**: Implement a feature to generate PDF documents containing comprehensive summaries and insights derived from the network analysis, including detected threats and anomalies.

6. **Create a Live IoT Environment for Demonstration and Alert System**: Set up a simulated IoT environment to demonstrate the functionality of the system, including controlling IoT devices like room lights, and integrate an alert system triggered by the detection of Mirai botnet activity.

CHAPTER 4

SYSTEM DEVELOPMENT

The chapter is to specify the software and hardware requirements of the system.It serves as the foundation for the development of the software providing a detailed description of what the software should do and how it should behave.

4.1 Software Requirements

4.1.1 Requirement Specification

Introduction

The purpose of this software requirement specification is to outline the functional and non-functional requirements for the development of Mirai Botnet Detection: ML with Live IoT Simulation.

Real-Time Packet Capture

- The system must be capable of capturing network packets in real-time from the connected network interface.
- It should ensure efficient packet capture to minimize latency and ensure accurate analysis.

Packet Analysis

- The tool must perform thorough analysis of captured packets, extracting essential metadata such as source IP, destination IP, source port, and destination port.
- It should process packet data to identify patterns and signatures associated with Mirai botnet activity.

Mirai Detection

- Implement machine learning algorithms or pattern recognition techniques to detect Mirai botnet behavior accurately.
- The detection mechanism should be robust and capable of distinguishing between benign and malicious network activity effectively.

Email Notifications

- Upon detecting Mirai botnet activity, the system should generate email notifications to alert designated recipients.
- Email notifications should include comprehensive packet summaries and analysis results for further investigation.

PCAP File Upload

- Allow users to upload pcap (packet capture) files for offline analysis and historical investigation.
- Implement file parsing functionality to extract packet data from uploaded pcap files and perform Mirai detection as with live packet capture.

IoT Network Simulation

- Develop a simulated IoT network environment to replicate typical IoT device interactions .

- Use the simulation to validate the detection capabilities of the system under various network conditions and attack scenarios.

Mobile Application Development

- Create a application using Flutter framework ..
- The application should provide remote device management features, including monitoring and controlling IoT devices connected to the network.

Automated Responses

- Implement automated response mechanisms upon Mirai botnet detection, such as isolating or quarantining compromised devices.
- Ensure that automated responses are configurable and can be tailored to the specific requirements of the network environment.

Alerting Mechanism

- Incorporate an alerting mechanism within the application interface to notify users of detected Mirai botnet activity.
- Alerts should be displayed in real-time, providing relevant information and actionable insights for prompt response and mitigation.

4.1.2 Non-Functional Requirements

Performance

- Ensure minimal impact on network performance.

Scalability

- Scale to accommodate growth in devices and traffic.
- Horizontal scalability for increased demands.

Reliability

- Operate reliably under varying conditions.
- Ensure high availability and fault tolerance.

Security

- Implement robust security measures.
- Encrypt data during transmission and storage.

Usability

- Design user-friendly interface.
- Provide clear visualizations for easy analysis.

4.1.3 Hardware Requirements

ESP8266

- Microcontroller board for IoT device management.
- Capable of Wi-Fi connectivity for network communication.

LED

- LED bulb compatible with IoT app for remote control.
- Enables switching light on/off based on network events.

Buzzer

- Audio signaling device for alert notifications.
- Triggers audible alerts in response to detected threats.

CHAPTER 5

SYSTEM DESIGN

This chapter defines the system architecture, usecase diagram and class diagram of the proposed system.These components serve as foundational elements in system design, providing a comprehensive overview of the structural organization, functional requirements, and object-oriented structure of the system. By detailing these aspects, the report aims to establish a clear understanding of the system's design principles, facilitating effective communication and guiding the development process.

5.1 System Architecture

The Figure 5.1 shows the the system architecture.It is the low-level design of the system which outlines the structure and components of the Mirai Detection tool. It provides an abstract view of how different modules or components interact with each other to achieve the system's functionality.

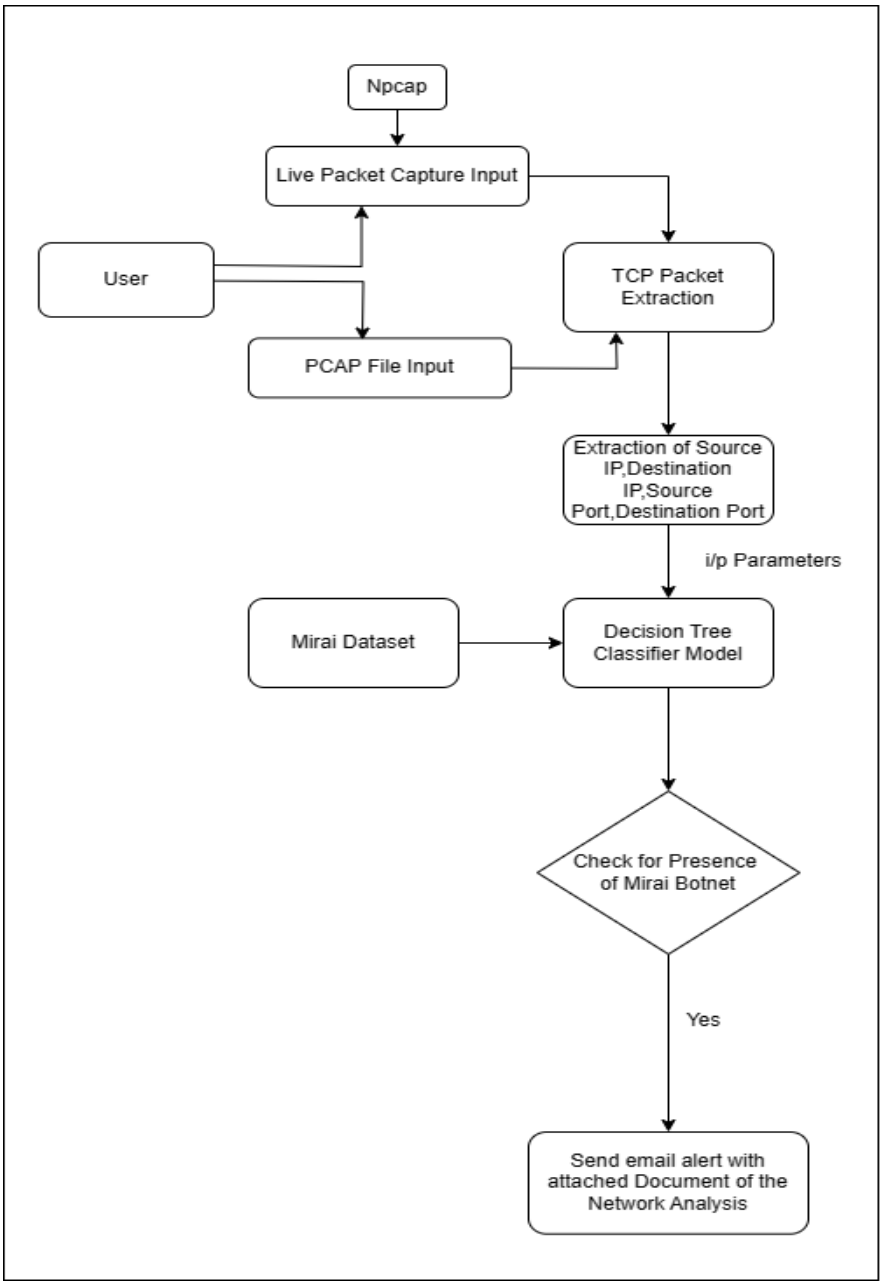


Figure 5.1: System Architecture

5.2 Class Diagram

The class diagram depicted in the Figure 5.3 shows the class diagram with the various functionalities which have various attributes and methods associated with the Mirai Detection System.The diagram showcases the overall design of the system, highlighting the classes and their interactions. It helps in understanding the system's

architecture, identifying key components, and facilitating communication among development teams.

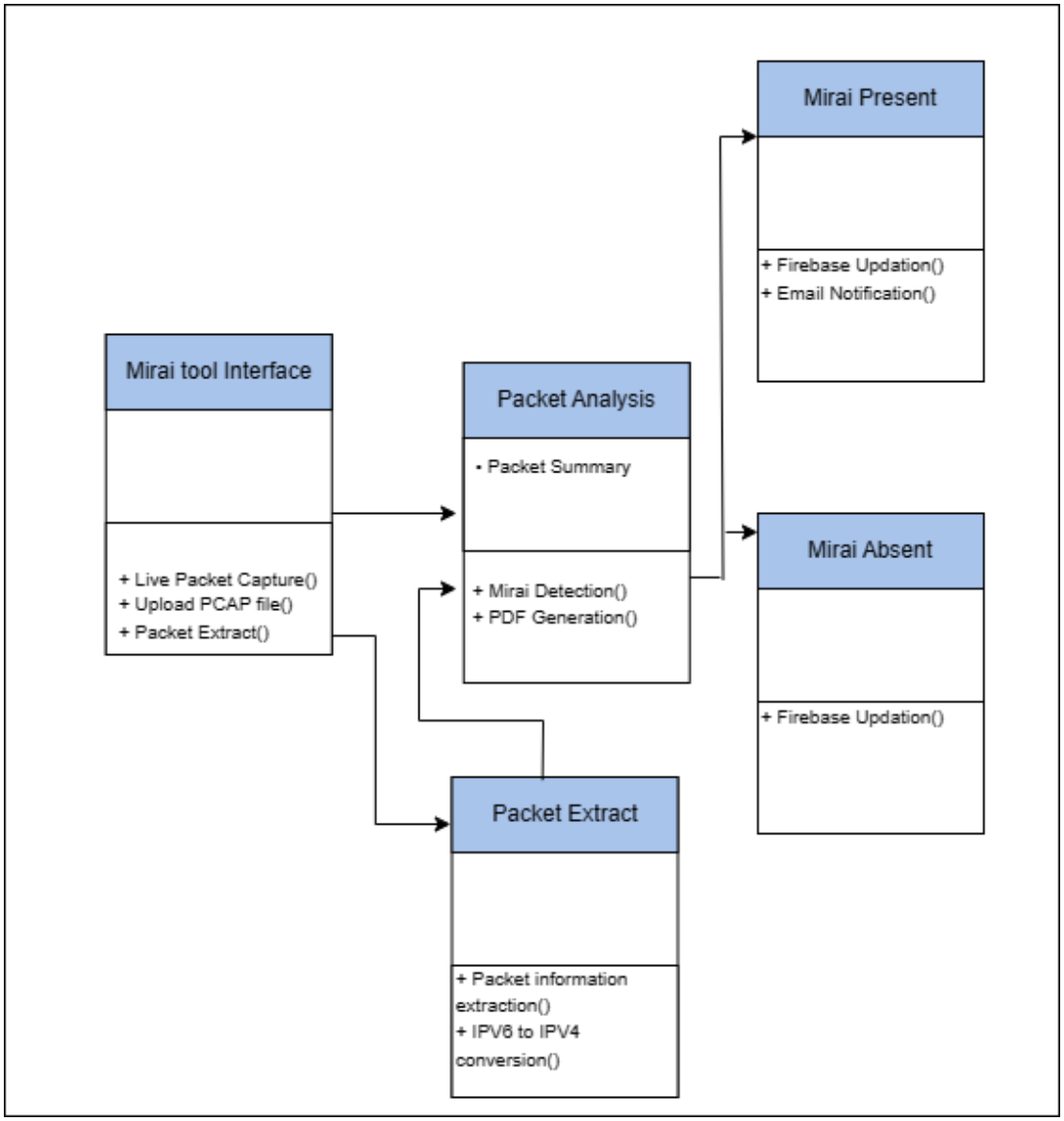


Figure 5.2: Class Diagram

5.3 Use Case Diagram

The use case diagram depicted in Figure 5.3 shows the relation between the user and the system. The figure shows the relation between the User as actor and the relationship between different functionality between the actor and app as use cases.This diagram is

helpful in capturing the system's intended functionality.

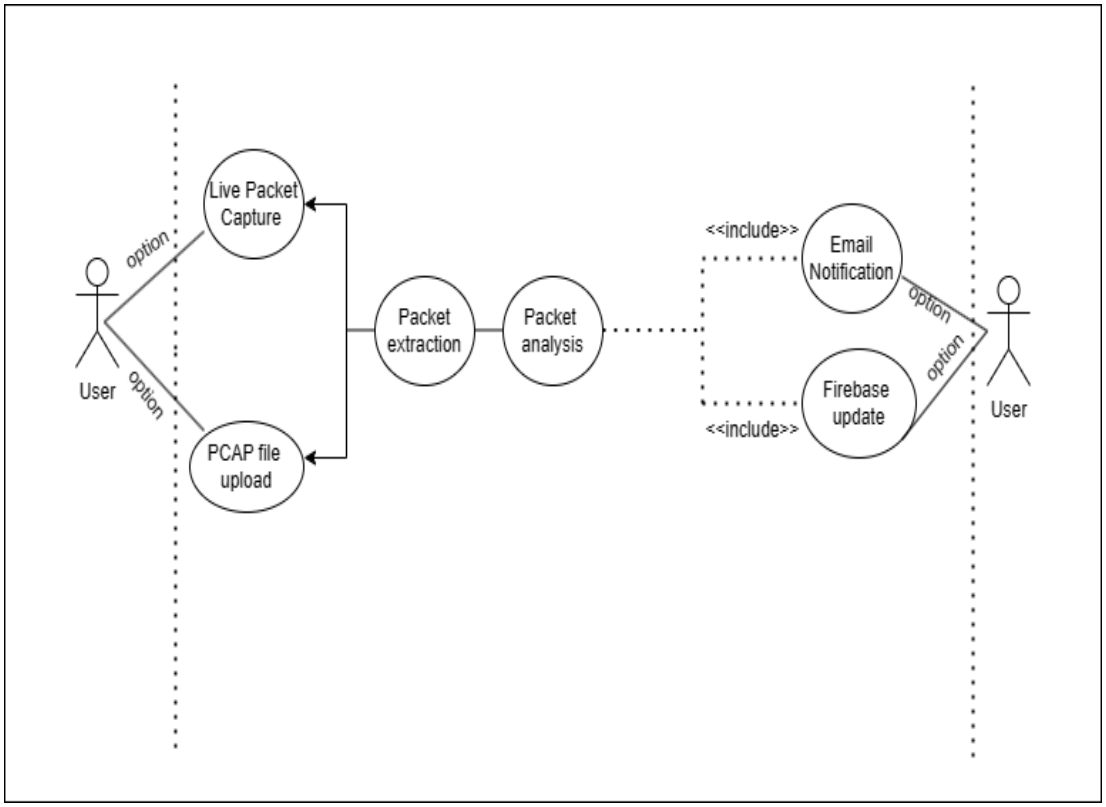


Figure 5.3: Use Case Diagram

CHAPTER 6

IMPLEMENTATION

6.1 IMPLEMENTATION TOOLS

6.1.1 Mirai Detection Tools

1. **Python**

   Python is a versatile and widely-used programming language known for its simplicity and readability. It offers a rich ecosystem of libraries and frameworks that make it suitable for various types of software development, including web development.

2. **Npcap**

   Npcap is a packet sniffing library and network capture utility designed for Windows systems. It extends WinPcap, offering enhanced functionality and support for modern Windows versions. With features like kernel-level packet filtering and injection, it ensures efficient packet handling and manipulation.

3. **SMTP**

   SMTP, or Simple Mail Transfer Protocol, facilitates the transmission of emails between servers and clients. It operates on port 25 and employs a client-

server architecture. SMTP servers handle outgoing mail, while email clients or other mail transfer agents interact with these servers to send messages. SMTP supports essential features like authentication, ensuring secure and efficient email delivery.

6.1.2 Mobile App Development Tools

1. **Flutter**

   Flutter is a UI toolkit developed by Google for building natively compiled applications for mobile, web, and desktop from a single codebase. It provides a framework for building user interfaces using a reactive programming paradigm and a rich set of pre-built UI components.

2. **Firebase**

   Firebase is a platform developed by Google that provides a variety of services and tools to help developers build, improve, and grow their applications. It offers services such as real-time database, authentication, cloud storage, hosting, analytics, and more, which can be integrated into mobile and web applications to add functionality and improve user experience.

3. **Android Studio**

   Android Studio is the official integrated development environment (IDE) for Android app development, provided by Google. It offers tools for writing, debugging, and testing Android applications, as well as features for managing project files, dependencies, and resources.

6.1.3 Hardware Components and Modules

1. **NodeMCU**

   The NodeMCU is a versatile Wi-Fi module commonly used in IoT projects. It allows devices to connect to the internet and communicate wireless, making it ideal for applications such as home automation, sensor monitoring, and remote control.

2. **LED**

   LED or Light Emitting Diode are semiconductor devices that emit light when an electric current passes through them. They are widely used for lighting purposes in IoT devices due to their energy efficiency and controllable properties.

3. **Piezo Buzzer**

   A piezo buzzer is an electronic device that generates sound when voltage is applied to it. It operates by rapidly vibrating a piezoelectric crystal to produce audible frequencies. Piezo buzzers are commonly used in alarm systems, electronic devices, and IoT applications for generating alerts, notifications, and alarms due to their simplicity, low cost, and efficiency.

4. **Arduino IDE**

   The Arduino Integrated Development Environment (IDE) is a software platform used for programming and uploading code to Arduino micro-controllers. It provides a user-friendly interface for writing, compiling, and uploading code, making it accessible to beginners and experienced developers alike.

6.2 RESULTS AND DISCUSSION

The project aimed to develop a warning system for detecting the Mirai Botnet. All objectives outlined in the scope and objectives section were successfully achieved. The subsequent section displays various interfaces and corresponding results for each functionality.

6.2.1 Algorithm Accuracy

The Decision-Tree Classifier outperformed five other algorithms in the Mirai Detection system, achieving the highest level of accuracy at 100%, as illustrated in Figure 6.1.

Technique	Wrong Predictions	Accuracy
K - Nearest Neighbour Classification	12	99.971%
Support Vector Machine Classification	20	95.215%
Kernel Support Vector Machine Classification	16	96.2085
Decision Tree Classification	0	100%
Random Forest Classification	5	98.845%

Figure 6.1: Accuracy of Decision Tree Classifier against other algorithms

6.2.2 Mirai Detection Tool

The Mirai detection Tool interface is shown in the Figure 6.2 with 2 buttons Live Packet Capture and Upload PCAP files.The Packet summary results with instances of Mirai and without Mirai is shown in Figure 6.3,additionally the Email notification sent with attached Packet Summary is shown in the Figure 6.4.

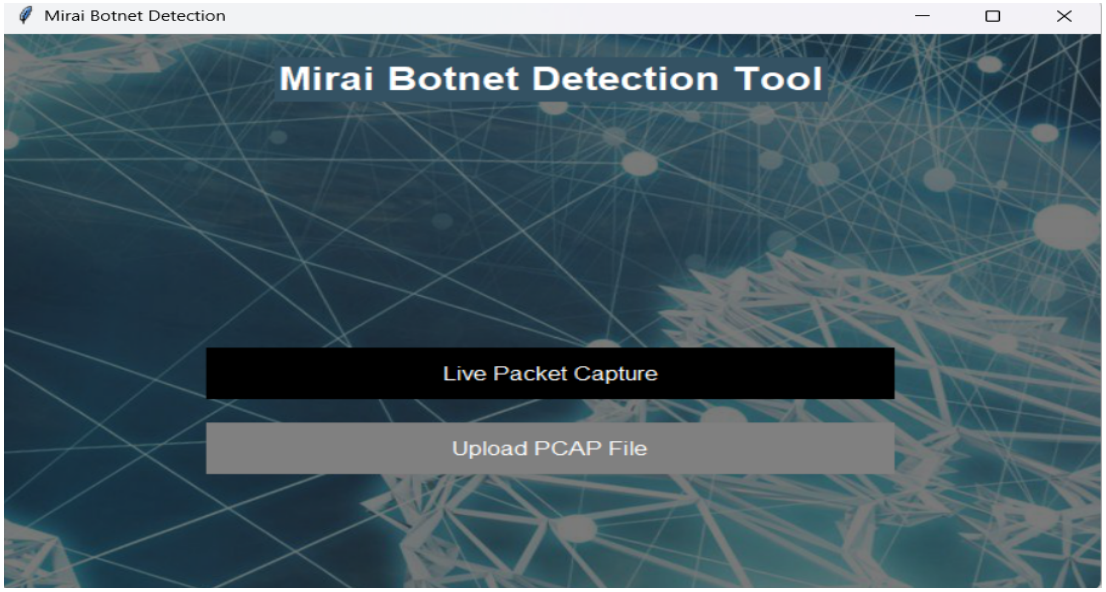


Figure 6.2: Mirai Detection main page

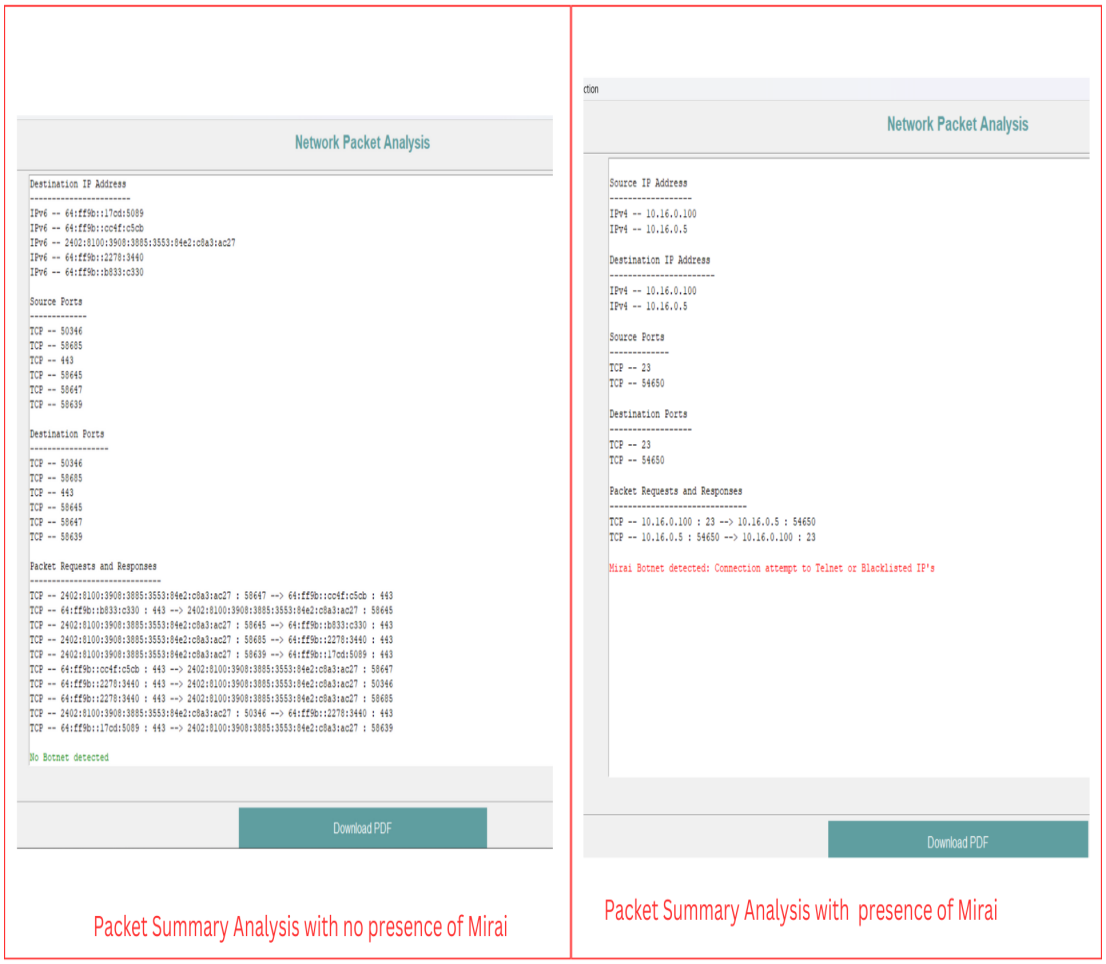


Figure 6.3: Packet Analysis

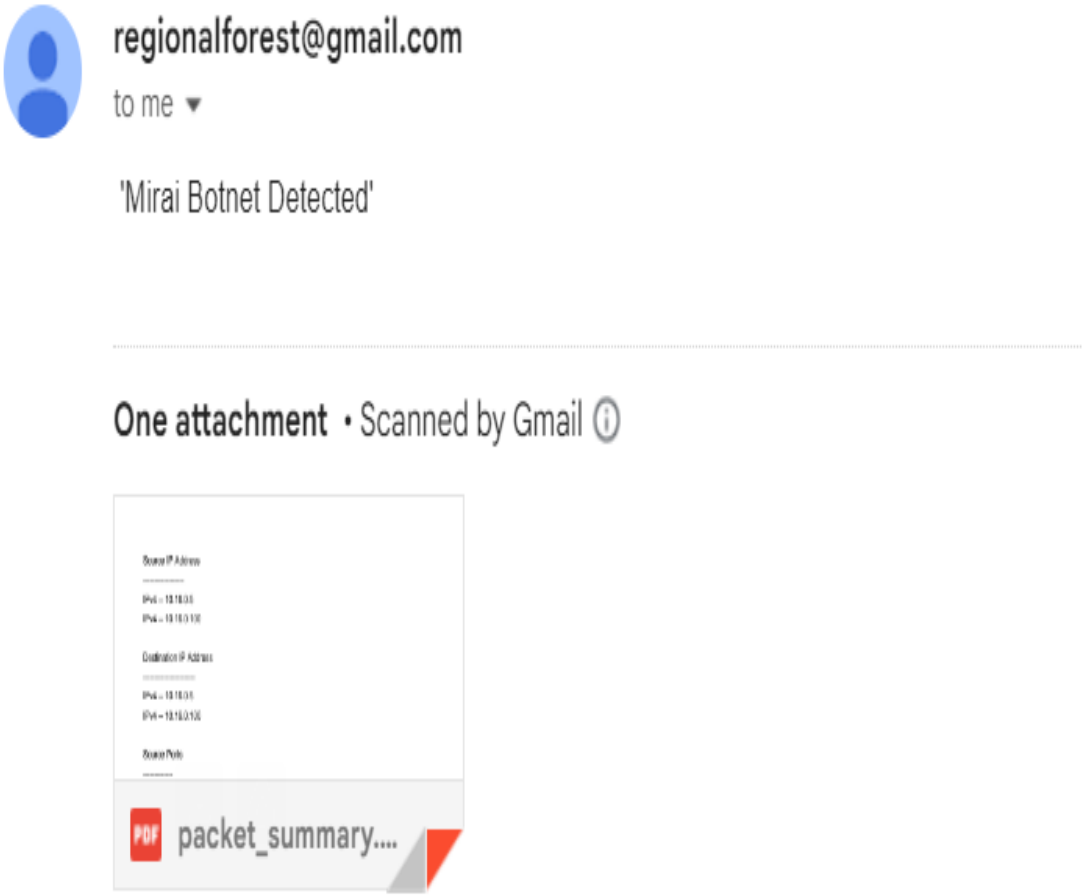


Figure 6.4: Email Notification

6.2.3 Realtime IoT App

The real-time system is implemented within an emulated Android application. The design of the application is not included in the system design as it serves primarily as a prototype to demonstrate the integration of Mirai alert functionality into an IoT system. This prototype does not represent an ideal system, and the requirements for IoT simulation at an industrial level may vary significantly.

The Android application is developed using Flutter and is deployed as a simulated Android application due to current limitations in Flutter versions, which do not support

Firebase Realtime Database via web apps. The application comprises a main app page, registration page, and login page, as depicted in Figure 6.5.

The application communicates with the IoT endpoints via Firebase. Although packet transfer in normal Firebase communication is encrypted, it is assumed that users are provisioned with authoritative access over their accounts to monitor Firebase communication and are permitted to sniff packets entering and leaving their Firebase Realtime Database.

The Firebase backend is connected to a microcontroller, where each button press controls the lighting of a bulb. Upon detection of Mirai activity, the bulb is set to an off state, and a buzzer beep is activated, persisting until the power supply to the microcontroller is interrupted. Additionally, the application displays an alert message, and normal control of the bulb can only be reinstated when Mirai activity is no longer detected on the network. The layout of the IoT app page instances is illustrated in Figure 6.6, and the circuit diagram for the bulb and buzzer is provided, with a LED used in lieu of a commercial bulb for prototyping purposes as presented in Figure 6.7

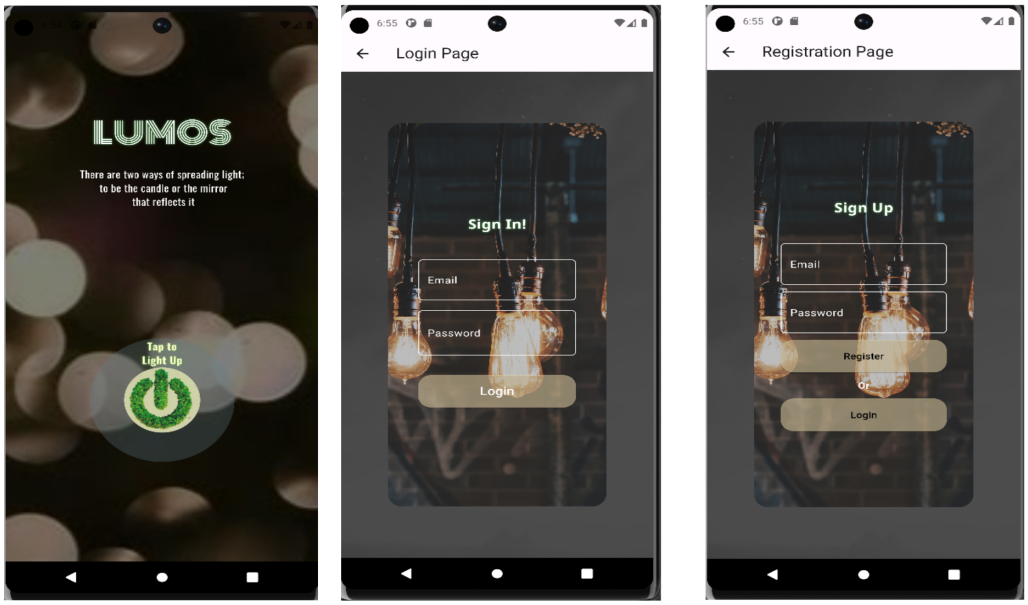
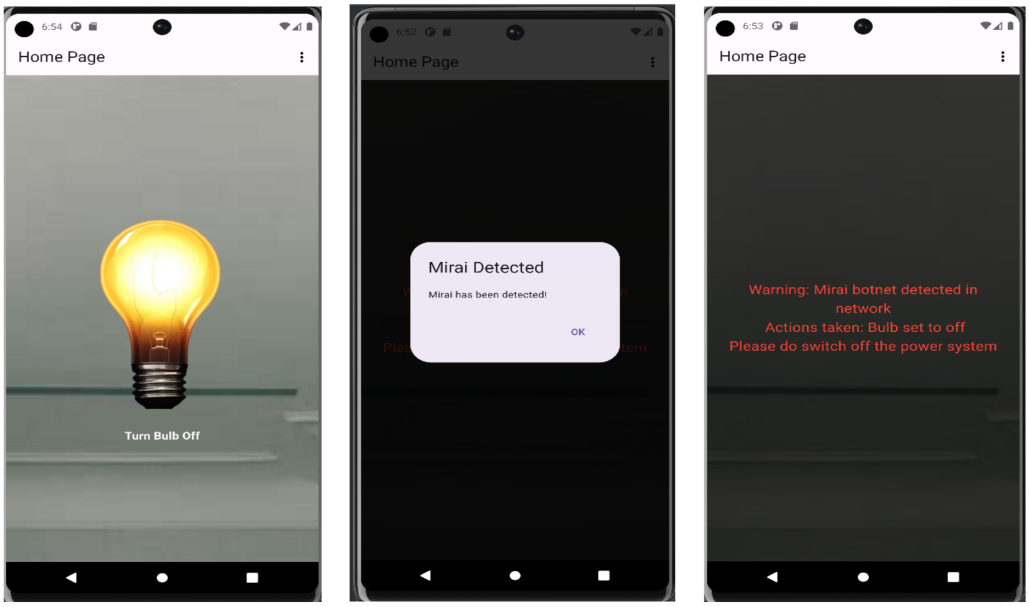


Figure 6.5: Authentication Pages



IoT page showing (1) normal mode ,(2) Alert mode ,(3) Threat mitigation in Comprimised mode

Figure 6.6: Home Page

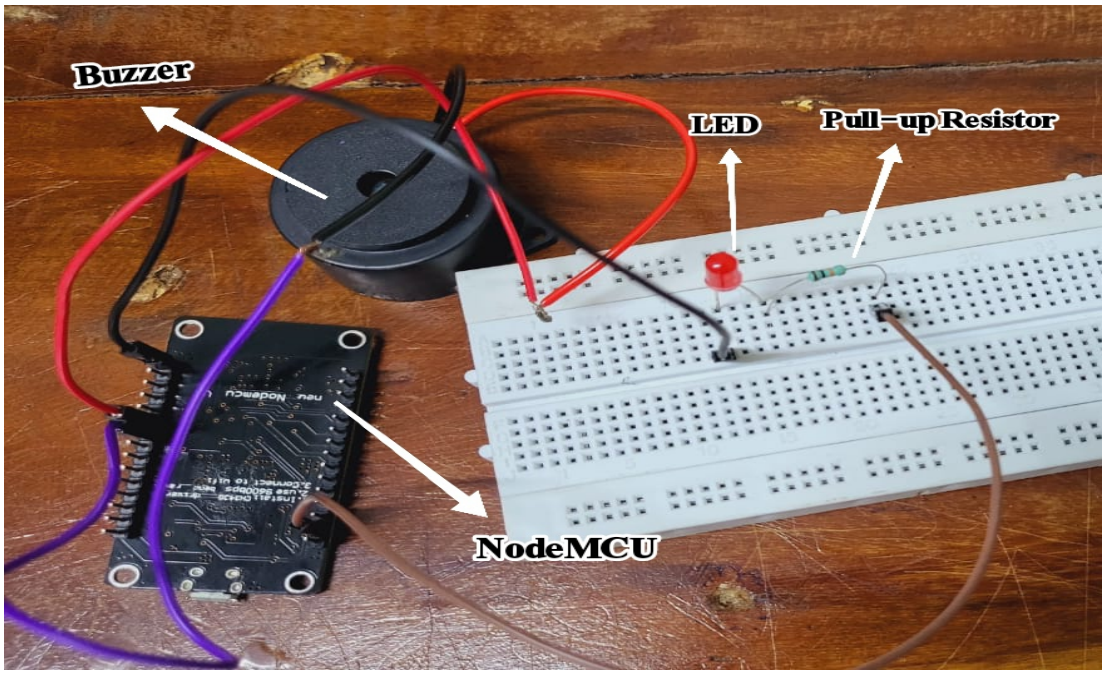


Figure 6.7: Circuit

CHAPTER 7

TESTING

7.1 Functionality Testing

Separate test cases are designed and tested for all the functional modules defines in the software requirement section.

- Verified accuracy of live network analysis.
- Verified Assess for PCAP file analysis functionality.
- Tested integration of machine learning model.
- Validated PDF document generation.
- Evaluated IoT environment simulation.

7.2 UI Testing

The system allows only registered users to use the interface in IoT Control App.The App has 4 pages Home Page,Sign In and Sign Up Page and App Page.The seamless integration of Firebase realtime databse with UI is ensured.The package tool has 2 pages for choose of opting Live Analysis or Captured Analysis.

- Verified user authentication.

- Verified the interface page and the fetched detailed results page.
- Verified the smooth rendering of UI change corresponding to actuation of led and buzzer.
- Verified the packet analysis results are rendered seamlessly without fault.

7.3 Integration Testing

The testing is done after all the modules of the system are completely developed, tested individually, and integrated. All the bugs during the development and testing are fixed and verified. Integration testing included the testing of the back end and interaction with the mobile app interface.

- Verified interaction between system components.
- Tested communication between modules.
- Ensured seamless integration of external interfaces.

CHAPTER 8

CONCLUSION

The development of the Mirai IoT botnet detection system represents a significant advancement in the field of network security and IoT device protection. Through the integration of live packet capture, machine learning algorithms, and real-time analysis, the system demonstrates its capability to effectively identify and mitigate threats posed by the Mirai botnet. By analyzing key packet information such as source and destination IPs, ports, and employing a trained ML model specifically targeting Mirai botnet patterns, the system ensures a robust defense mechanism against potential attacks.Moreover, the system's ability to generate detailed PDF reports and provide real-time alerts enhances its usability and practicality for security professionals and network administrators. The inclusion of an IoT environment simulation further strengthens its applicability by showcasing how the system can actively respond to detected threats, thereby safeguarding IoT devices and networks from potential harm.The successful implementation and testing of the system validate its functionality, reliability, and effectiveness in detecting and combating the Mirai botnet.To summarize, the Mirai IoT botnet detection system represents a significant step forward in bolstering the security posture of IoT networks. Its comprehensive features, robust architecture, and proactive defense mechanisms position it as a valuable tool in the ongoing battle against malicious cyber activities, ultimately contributing to a safer and more secure digital ecosystem.

APPENDIX

8.1 Definitions

1. **Wi-Fi**: Wi-Fi is short for ”Wireless Fidelity.” In smart farming, Wi-Fi is used to enable wireless communication between devices and sensors, allowing for data transmission and remote monitoring.

2. **IoT**: IoT stands for the ”Internet of Things.” In smart farming, IoT refers to the network of interconnected devices, sensors, and equipment that collect and exchange data to optimize farm operations and decision-making.

3. **ML**:ML stands for ”Machine Learning” .It’s an Automating learning from data to make predictions without explicit programming.In the project’s context it’s used for Crop Prediction and Cost Estimation.

4. **NodeMCU**:NodeMCU stands for Node Microcontroller Unit.Open-source firmware for ESP8266, enabling IoT prototyping.This micro-controller is used as the sole source for programming all IoT components.

5. **PCAP**:Packet Capture, which refers to the process of capturing and storing network traffic packets that pass through a computer network interface.

6. **PDF**:PDF stands for Portable Document Format, a file format developed by Adobe Systems that captures the layout of a document, including text, images, fonts, and formatting, in a fixed-layout format that can be viewed consistently across different platforms and devices.

7. **DDoS**:Distributed Denial of Service is a cyberattack where multiple compromised systems flood a target with excessive traffic, disrupting its normal operation and rendering it inaccessible to legitimate users.

8. **IP**:Internet Protocol is a network protocol responsible for addressing and routing packets of data across networks.

9. **IPV4**:Internet Protocol version 4 is the fourth version of the Internet Protocol, which identifies and routes network traffic across the internet, primarily using 32-bit addresses.

10. **IPV6**:Internet Protocol version 6 is the most recent version of the Internet Protocol, designed to succeed IPv4, offering a much larger address space using 128-bit addresses.

11. **TCP**:Transmission Control Protocol is an internet protocol responsible for establishing and maintaining reliable, connection-oriented communication between networked devices by managing the transmission of data packets.

12. **SMTP**:Simple Mail Transfer Protocol is a standard protocol used for sending and receiving email messages over the internet.

REFERENCES

[1] A. Affinito, S. Zinno, G. Stanco, A. Botta, and G. Ventre, 'The evolution of Mirai botnet scans over a six-year period', *Journal of Information Security and Applications,* vol. 79, p. 103629, 2023.

[2] J. Kim, M. Shim, S. Hong, Y. Shin, and E. Choi, 'Intelligent Detection of IoT Botnets Using Machine Learning and Deep Learning',' *Applied Sciences*, vol. 10, no. 19, 2020.

[3] L. F. Sikos, 'Packet analysis for network forensics: A comprehensive survey', *Forensic Science International: Digital Investigation,* vol. 32, p. 200892, 2020.

[4] B. Charbuty and A. M. Abdulazeez, 'Classification Based on Decision Tree Algorithm for Machine Learning', *Journal of Applied Science and Technology Trends*, 2021.

[5] B Padmaja, E Krishna Rao Patro, Sankeerth Mahurkar, Akhila G, 2021, Google Firebase based Modern IoT System Architecture, *INTERNATIONAL JOURNAL OF ENGINEERING RESEARCH TECHNOLOGY (IJERT) ICACT* ,2021 (Volume 09 – Issue 08),